

PAPER 1 / QUESTION 93

2224-6701 / original question 18

93. An object of mass m is attached to the end of a rope. The object is rotated in a horizontal circle at the end of the rope with an increasing speed. The rope breaks when the angular velocity ω is reached.

At what other combination of mass and angular velocity will this rope break?

	Mass	Angular velocity
A.	$3m$	$\frac{\omega}{2}$
B.	$4m$	$\frac{\omega}{4}$
C.	$3m$	$\frac{\omega}{4}$
D.	$4m$	$\frac{\omega}{2}$